Statement of Purpose

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My primary research interest lies in learning for natural language understanding; I intend to design systems and develop algorithms that address barriers to human-level understanding of natural language. My motivation comes from the curiosity about learning and developing intelligence; human's ability to learn, teach and express ideas with *language* has always captivated me. As my background is in computer science, it has led me to study artificial intelligence and aim to build a system that is capable of advanced communication.

I have been fortunate to work as an undergraduate researcher, focusing on the task of reading the text and answering the question. I was motivated by the idea that question answering is not only the basic form of communication, but also a reasonable way to evaluate an ability to understand natural language. The result of my efforts have been published in ICLR'17 and ACL'17, and submitted to ICLR'18. The first paper explored synthetic reasoning as a preliminary step in modeling human reasoning. The other two works focused on tasks that require lexical and syntactic inferences from real documents. As a lead author, I showed that the task of question answering can benefit from large fine-grained supervision data by transfer learning across different tasks and domains. More recently, I worked on a variant of RNN which imitates a human's ability to skim the text; I showed that these variants could lead to an improvement in the speed of inference on text classification and question answering. Currently, I am in the beginning of another project as a lead author, aiming to address another challenge in the task of question answering.

These research experiences have helped me in two ways. First, they have enabled me to build a solid foundation and skills as a researcher; I have learned how to define problems, how to design methodologies, how to implement ideas, and how to run experiments efficiently and reproducibly. Second, they have led to approaches to my future studies, as will be described below.

First, I am interested in well-defining the task, as I believe that challenges can be better addressed when we have a clear definition of what we desire to solve. I was inspired by the contribution of SQuAD to the recent enhancement of a machine's ability to read and comprehend. In particular, I believe that the span-based answer in SQuAD, which has a straight-forward evaluation metric, has significantly fostered the machine's performance. As next steps to the span-based answer, I am interested in the tasks of 'answering questions via natural sentences' or 'asking good questions'. However, these problems have a severe limitation in evaluation because they mostly deal with *sentence*. In fact, during my past project, I have struggled to deal with answer sentence generation, and faced the fundamental questions – (i) what is a good answer sentence, and (ii) is n-gram precision the best way to evaluate sentences? Although I have not yet found the clear answer to these questions, I would like to explore solutions to these fundamental challenges, as a primary step to tackle the problem of sentence generation.

Second, I intend to explore learning algorithms that can model the complex behavior of human. For example, I believe that humans perform many different tasks based on common language features, which has led me to be intrigued by representation learning, multi-task learning and transfer learning. In particular, my past work has shown that fine-grained supervision data leads to an effective learning of common language features such as lexical information, thus, it can benefit other tasks through transfer learning. I think more different tasks can benefit each other; for instance, I hope to work on joint learning of asking questions and answering questions, since learning to ask good questions can contribute to answering questions and vice versa. Such work would be essential to building a dialogue system, which is also one of my interests. Another topic of my interest is to develop algorithms for large-scale problems such as open-domain question answering. I have handled this topic with a hierarchical structure, have encountered limitations, and intend to overcome them in two ways. First, I hope to combine works from information retrieval and neural models in a better way – for example, by allowing neural models to dynamically retrieve information from data. Second, I would like to design a lightweight or cost-efficient model; in fact, it has been a motivation for my subsequent work that enhanced the speed of inference by allowing RNN to skim the less important words when reading the text. I would like to make neural models further lightweight in the future.

Lastly, I have a special interest in the intersection of natural language processing and computer vision; I am intrigued by problems which require interactions between language and visual data. I aim (i) to well-define tasks which require inference and reasoning on both modalities, and (ii) to design systems and algorithms to handle multimodal features. In particular, working as an intern in both natural language processing group and computer vision group, I noticed that language features and visual features behave very differently, but are often simply treated in a same manner. This observation has led to my interest in modeling the combination or complex interaction of language space and visual space with more effective methods. For this objective, I have tried to make contribution on multimodal problems by working on a project of image caption generation, presenting a seminar on visual question answering, and open-sourcing codes on multimodal problems.

Along with my plans to further the study of aforementioned ideas, my next career goal is to lead a research group as a professor. I have been particularly motivated by my advisors and mentors who have enabled me to reach my full potential. As they have done, I want to contribute to the research community not only through my own research but also by educating students for the next generation of research.

To accomplish my goals, I would like to pursue a Ph.D. at the University of Washington. Specifically, I look forward to working with Professor Hannaneh Hajishirzi and Professor Ali Farhadi. Working with Professor Hajishirzi and Professor Farhadi at the University of Washington has already provided me with the most fitting environment to conduct my research. In addition, their goals in research including (i) language understanding with a focus on question answering, reading comprehension and reasoning, and (ii) interactions between language and vision perfectly align with my research interest. In this respect, I am firmly convinced that the University of Washington is the ideal place for my career goal, and I am highly motivated to do my utmost to contribute to the University of Washington's program as a competitive researcher.